

Difference-in-Differences Analysis of Wastewater Treatment Upgrade Impact on Antibiotic Resistance Genes

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Abstract - Wastewater treatment is a major conduit for antibiotic resistance gene (ARG) dissemination, but evidence linking upgrades in treatment to ARG reduction remains limited. A difference-in-differences (DiD) experimental design was used to evaluate the effect of wastewater treatment infrastructure upgrades on total ARG abundance across forty wastewater samples from eight treatment plants in six countries. Three specific hypotheses were defined: H1, treatment upgrades are effective in reducing ARG abundance in treated plants in comparison to controls; H2, the treatment effect is robust to adjustment for seasonality, plant heterogeneity, and sequencing depth; H3, the magnitude of the effect varies by country due to systematic differences in infrastructure and treatment protocols. Contrary to our hypothesis, treated plants did not show significantly greater ARG reductions compared to controls. These findings suggest that wastewater treatment upgrades as implemented in the studied plants were not proven to reduce ARG levels, though small sample size warrants cautious interpretation.

Keywords: Wastewater treatment, antibiotic resistance, difference-in-differences,

1. Introduction

Responsible for an estimated 1.27 million deaths in 2019 and contributing to nearly 5 million deaths annually, antibiotic resistance is a growing global health crisis [1]. Although much attention has been focused on the clinical use of antibiotics, environmental reservoirs, especially wastewater treatment plants play a critical role in the dissemination of antibiotic resistance genes (ARGs) [2]. Wastewater treatment plants receive effluents from hospitals, households, industries, and more, creating hotspots where horizontal transfer of ARGs is promoted [3].

While infrastructure upgrades to wastewater treatment plants are widely implemented to improve effluent quality, evidence linking specific upgrades to meaningful reductions in ARG abundance remains limited and largely observational. Most existing studies employ before-after or cross-sectional comparisons, which are unable to isolate treatment effects from secular trends, seasonality, or time-invariant factors [4].

To address this gap, a natural experiment comparing plants that received infrastructure upgrades against control plants that did not was established. Using a difference-in-difference (DiD) design, we estimated the effects of three specific treatment upgrades on total ARG abundance. The DiD approach compares the change in ARG levels over time between treated and control groups, thereby removing time-invariant factors and common temporal trends [5].

2. Methods

2.1 Design and Sample Selection

We used a DiD natural experiment to evaluate whether wastewater treatment upgrades reduce ARG abundance. Forty wastewater samples were collected from eight treatment plants across six countries

(China, Italy, France, Portugal, UK, Netherlands). Each upgraded plant was matched with a control plant with no upgrades in the same time window. Treated plants received one of three upgrade types: Constructed Wetland (CW), where water passes through planted substrate for the filtration and biological breakdown of matter; Granular Activated Carbon (GAC), where water flows through activated carbon columns and dissolved pollutants are absorbed; AOP line (ozonation + GAC), where ozone first chemically oxidizes contaminants, then GAC adsorbs remaining byproducts.

2.2 DiD Model and Outcome Measurement

3. Results

3.1 ARG Abundance Trends

Prior to upgrades, treated plants had lower median ARG abundance (0.05×10^{12}) relative to control plants (1.18×10^{12}). Following the upgrades, the ARG abundance of treated plants increased to 1.34×10^{12} while that of control plants decreased to 0.84×10^{12} . This pattern indicates that, directly contrary to expectations, treated plants experienced increased levels of ARGs while controls decreased.

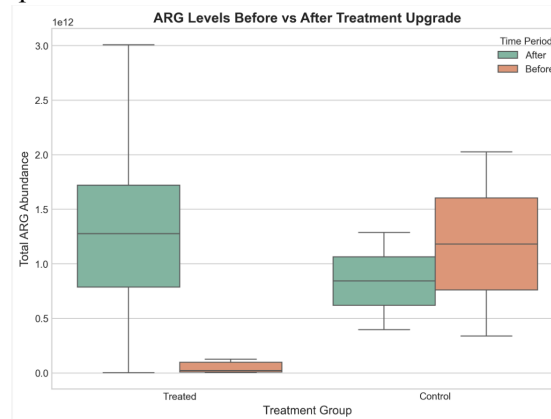


Figure 1. ARG levels before vs. after treatment upgrades by treatment group



Figure 2. Treatment vs. control trends in mean ARG abundance over time

3.2

The DiD analyses was conducted using the following model:

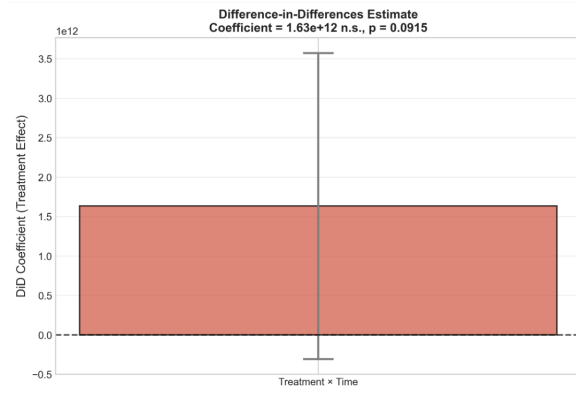


Figure 3. Difference-in-differences estimate

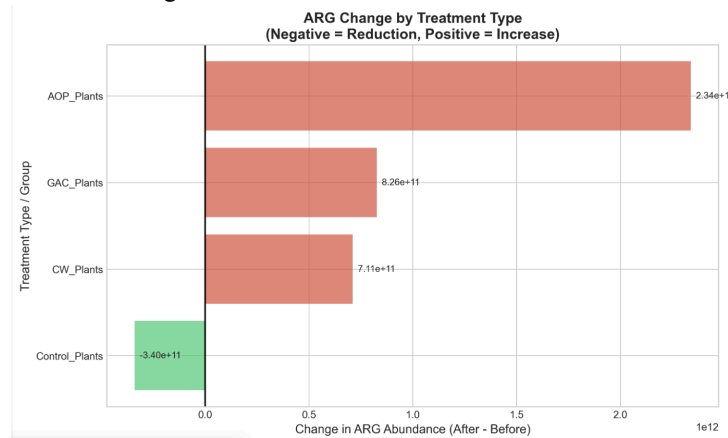


Figure 4. Change in ARG abundance (after - before) by treatment type. Negative values (green) indicate reduction while positive values (red) indicate increase.

4. Discussion

This study evaluated whether wastewater treatment infrastructure upgrades reduce ARG abundance using a difference-in-differences natural experiment across 40 samples from 8 plants in 6 countries. Contrary to our primary hypothesis (H1), evidence that treatment upgrades reduce ARG levels was not found. The DiD coefficient was positive ($\beta = 1.63 \times 10^{12}$) rather than negative, and the effect was not statistically significant ($p = 0.091$). The positive coefficient and non-significant p-value provide no statistical evidence that treatment upgrades reduce ARG abundance.

Several factors may explain this null finding. First, with a limited sample size, our study was underpowered to detect moderate effect sizes. Second, treated plants had substantially lower baseline ARG levels than controls (0.05×10^{12} vs. 1.18×10^{12}), leaving less room for improvement. Third, the three upgrade types (CW, GAC, AOP) all showed increases rather than decreases, suggesting these specific upgrades may not be effective for ARG reduction.

Country-level and treatment-type variation was observed, with all three upgrade types associated with ARG increases while controls decreased. This pattern raises the possibility that either upgrades themselves or unmeasured associated factors may have contributed to increased ARG levels.

5. Conclusion

Using a difference-in-differences natural experiment across 40 plant samples, this study evaluated whether wastewater treatment infrastructure upgrades had an effect in reducing ARG abundance. We found a positive but non-statistically significant treatment effect ($\beta = 1.63 \times 10^{12}$, 95% CI: -3.06×10^{11} to

3.57×10^{12} , $p = 0.091$). This provides no statistical evidence that CW, GAC, or AOP wastewater treatment upgrades reduce ARG abundance.

These findings have three primary implications. First, contrary to expectations, our analysis does not support the use of CW, GAC, and AOP line wastewater treatment upgrades as an effective intervention strategy to mitigate the environmental dissemination of ARGs. Second, the observed increase in all three upgrade types while controls decreased warrants further investigation. Third, they demonstrate the value of leveraging natural experiments and quasi-experimental models in environmental research.

Future research should prioritize long term follow-up to assess persistence of treatment effects and potential ARG rebound as well as larger sample sizes to achieve adequate statistical power.

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